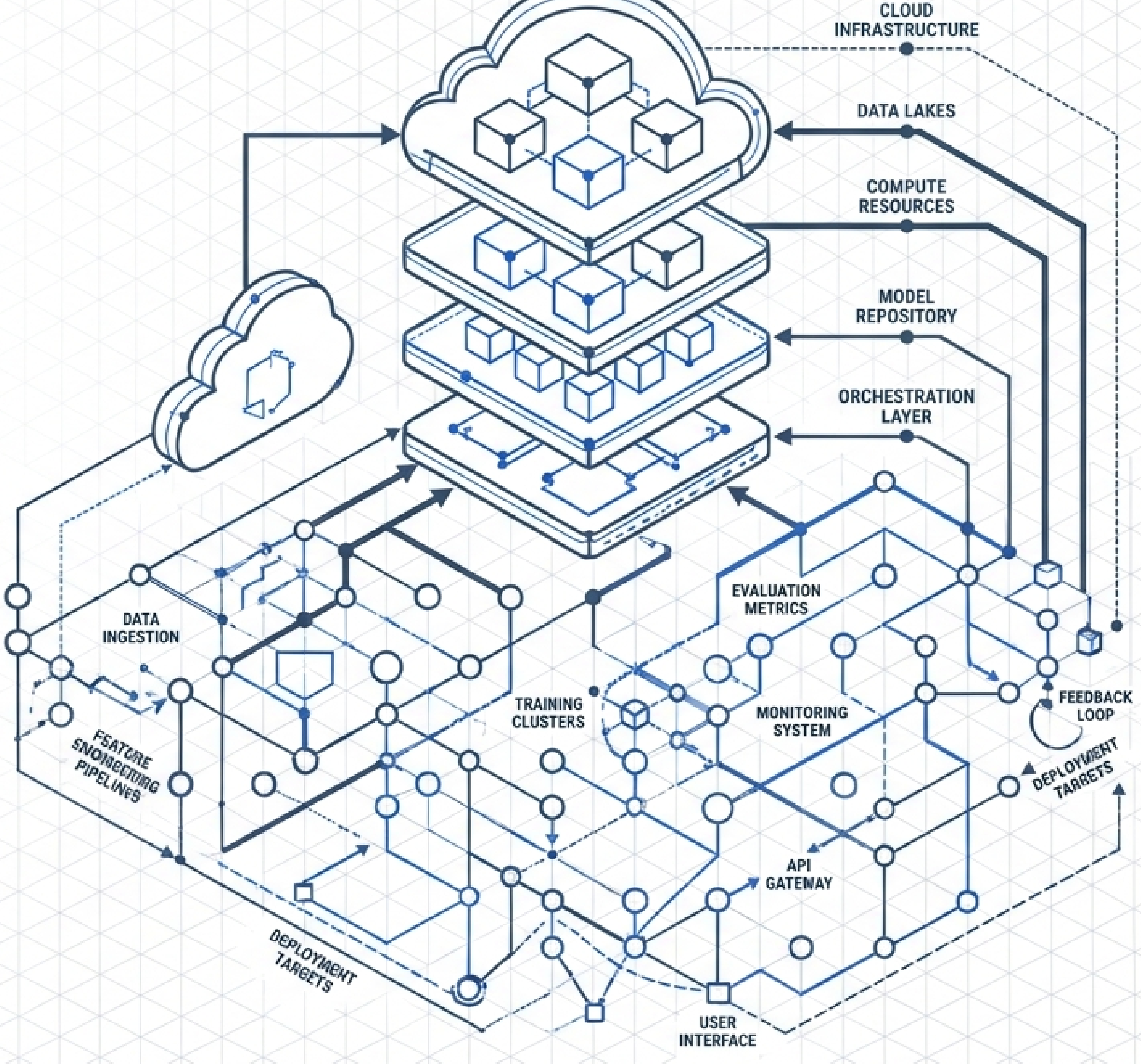
The Modern AI Engineering Blueprint

An End-to-End Playbook for Designing, Building, Evaluating, and Deploying AI Systems



Technical

Python, C++, Java/Scala. Mastery of linear algebra, probability, and optimization (gradient descent).

Frameworks

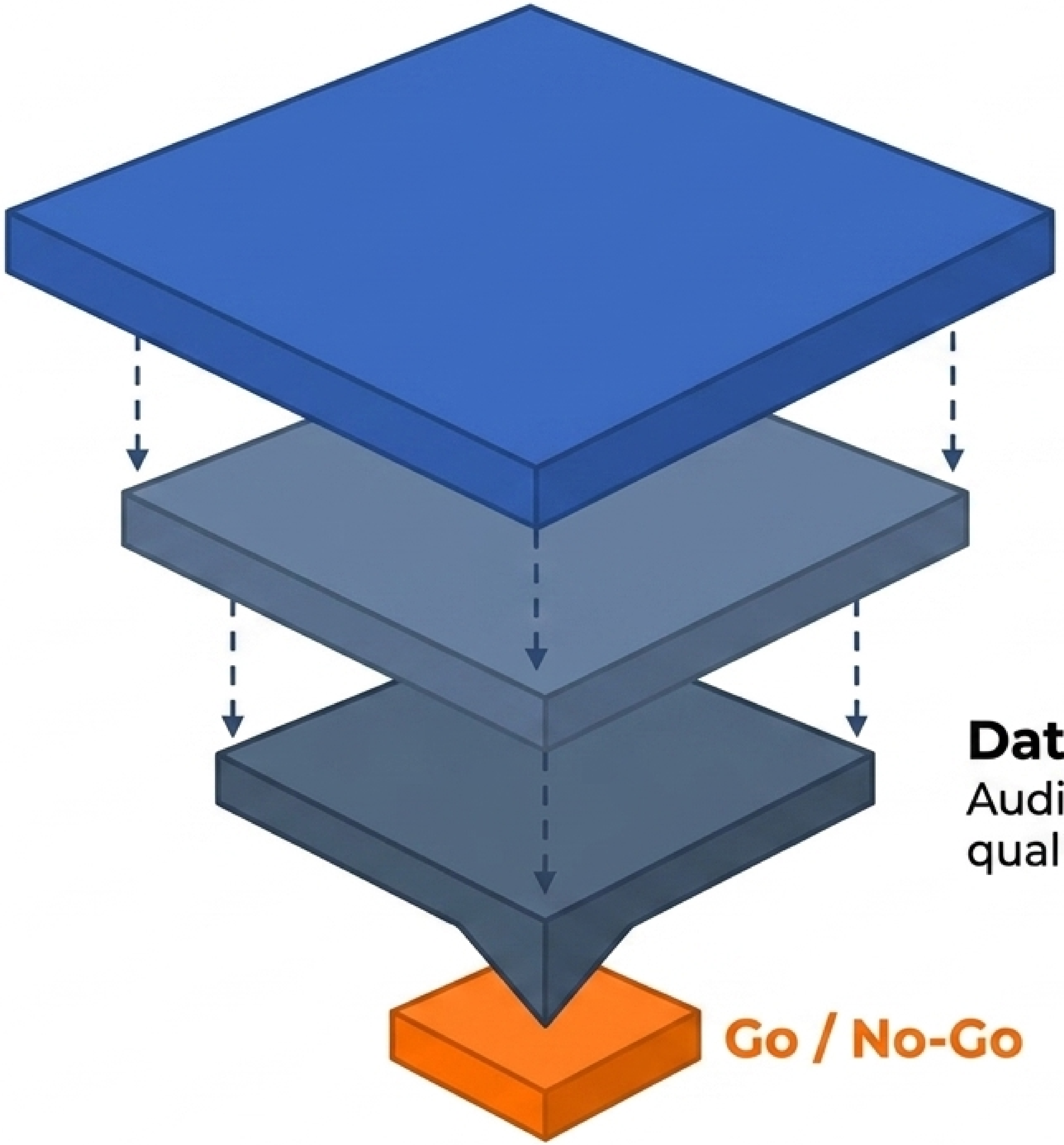
PyTorch, TensorFlow, Scikit-learn, Keras.

Data & DevOps

SQL/NoSQL, Hadoop/Spark, Docker, Kubernetes, CI/CD pipelines (Azure DevOps).

Domain & Soft Skills

Translating business goals into ML metrics, cross-functional collaboration, continuous learning.



Problem

Identify Pain & Gain. Calculate the Value-Difficulty Ratio at the activity cluster level to find the sweet spot.

Abilities (The 4 As)

Define required model capabilities: **Access** (retrieve info), **Analyze** (data patterns), **Automate** (routine tasks), or **Advance** (create new content).

Data

Audit domain-specific data for readiness, quality, and source integration feasibility.

```
"github.copilot.enable": {"*": false, "python": true}
"github.copilot.inlineSuggest.enable": false
"github.copilot.suggestion.length": 20
"github.copilot.advanced.editorContext": "currentLine"
"github.copilot.suggestionDelayMs": 500
```

Speed vs. Noise

Prevent noise in unsupported files and disable distracting inline overlays.

Pacing & Accuracy

Limit max lines per suggestion and enforce a 500ms delay to allow manual typing flow.

Context Privacy

Restrict data sent to Copilot to avoid leaking full file context.

Bronze Layer (Raw Ingestion)

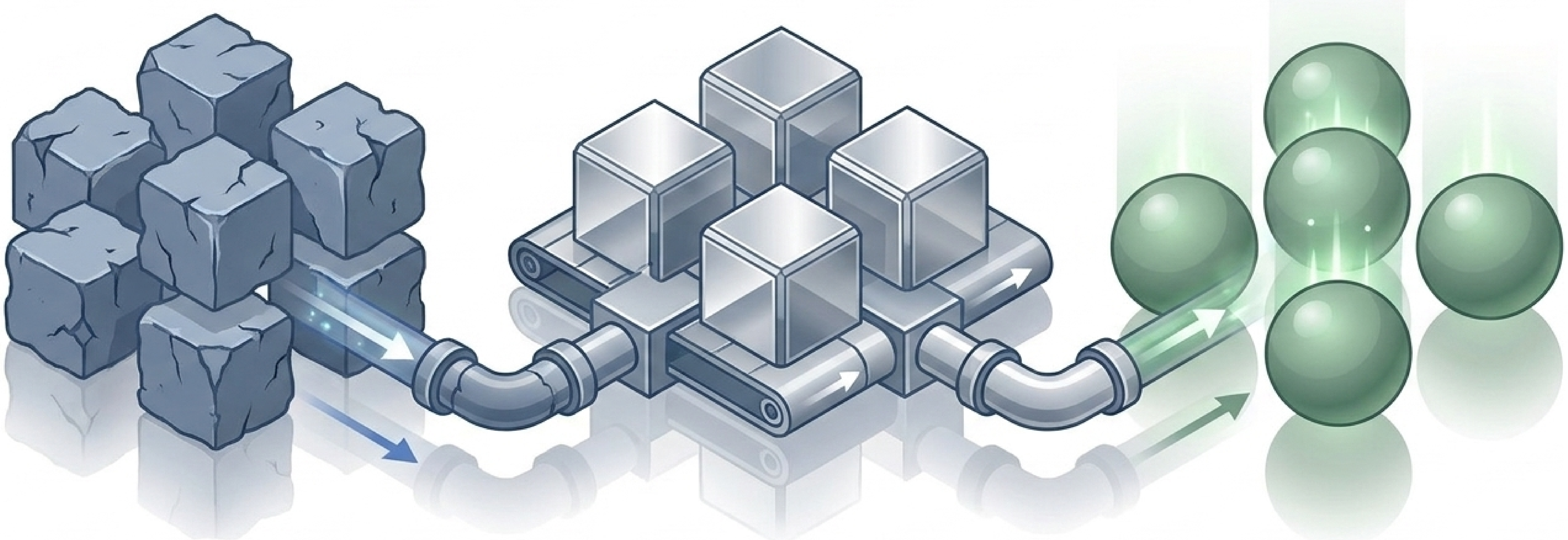
External data ingested via Azure Data Factory. Appends basic metadata (ingest_ts).

Silver Layer (Cleaned & Filtered)

Processed via Azure Databricks. Handles missing values (dropna), normalizes features (feature_1 / 100.0), and applies log transformations (F.log1p).

Gold Layer (ML-Ready)

Aggregated, final feature sets ready for semantic models, PowerBI, and ML training (gold.ml_training_data).



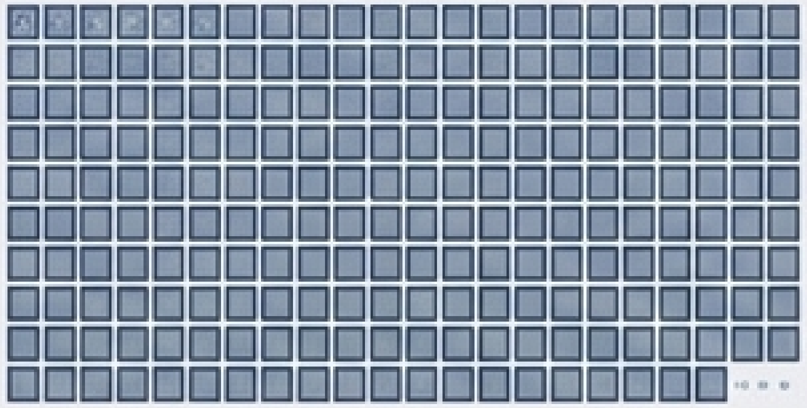
IBM taught me tokenization.

Word-Based (NLTK/spaCy)

IBM taught me tokenization.

["IBM"] ["taught"] ["me"] ["tokenization."]

Massive Vocabulary Array (Distinct IDs)



Maps whole words. Creates massive vocabularies; misses semantic links (e.g., treating **Unicorn** and **Unicorns** as completely distinct IDs).

Character-Based

IBM taught me tokenization.

['I'] ['B'] ['M'] [' '] ['t'] ['a'] ['u'] ['g'] ['h'] ['t'] [' '] ['m'] ['e'] [' '] ['t'] ['o'] ['k'] ['e'] ['n'] ['i'] ['z'] ['a'] ['t'] ['i'] ['o'] ['n'] ['.']

Small Vocabulary Array (Character IDs)

Maps single letters. Small vocabulary, no Out-Of-Vocabulary (OOV) tokens, but loses all inherent semantic meaning and inflates sequence length.

Subword-Based (WordPiece/SentencePiece)

IBM taught me tokenization.

["IBM"] ["taught"] ["me"] ["token"] ["##ization."]

The transformer standard. Splits rare words into meaningful subwords.

BERT: ['token', '##ization']
XLNet: ['■token', 'ization']

Special tokens:
<bos>, <eos>, <pad>, <unk>

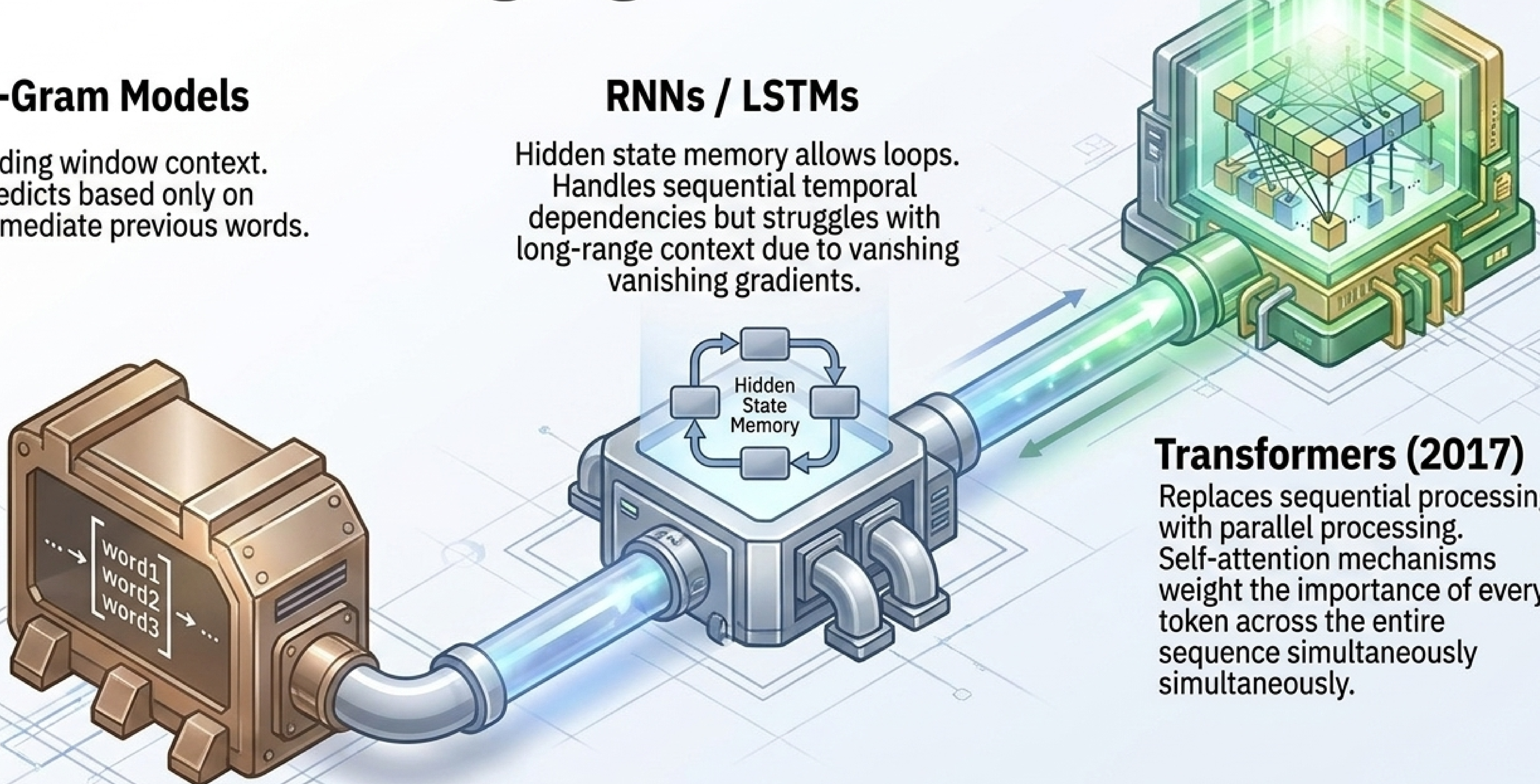
Evolution of Language Models.

N-Gram Models

Sliding window context.
Predicts based only on immediate previous words.

RNNs / LSTMs

Hidden state memory allows loops. Handles sequential temporal dependencies but struggles with long-range context due to vanshing vanishing gradients.

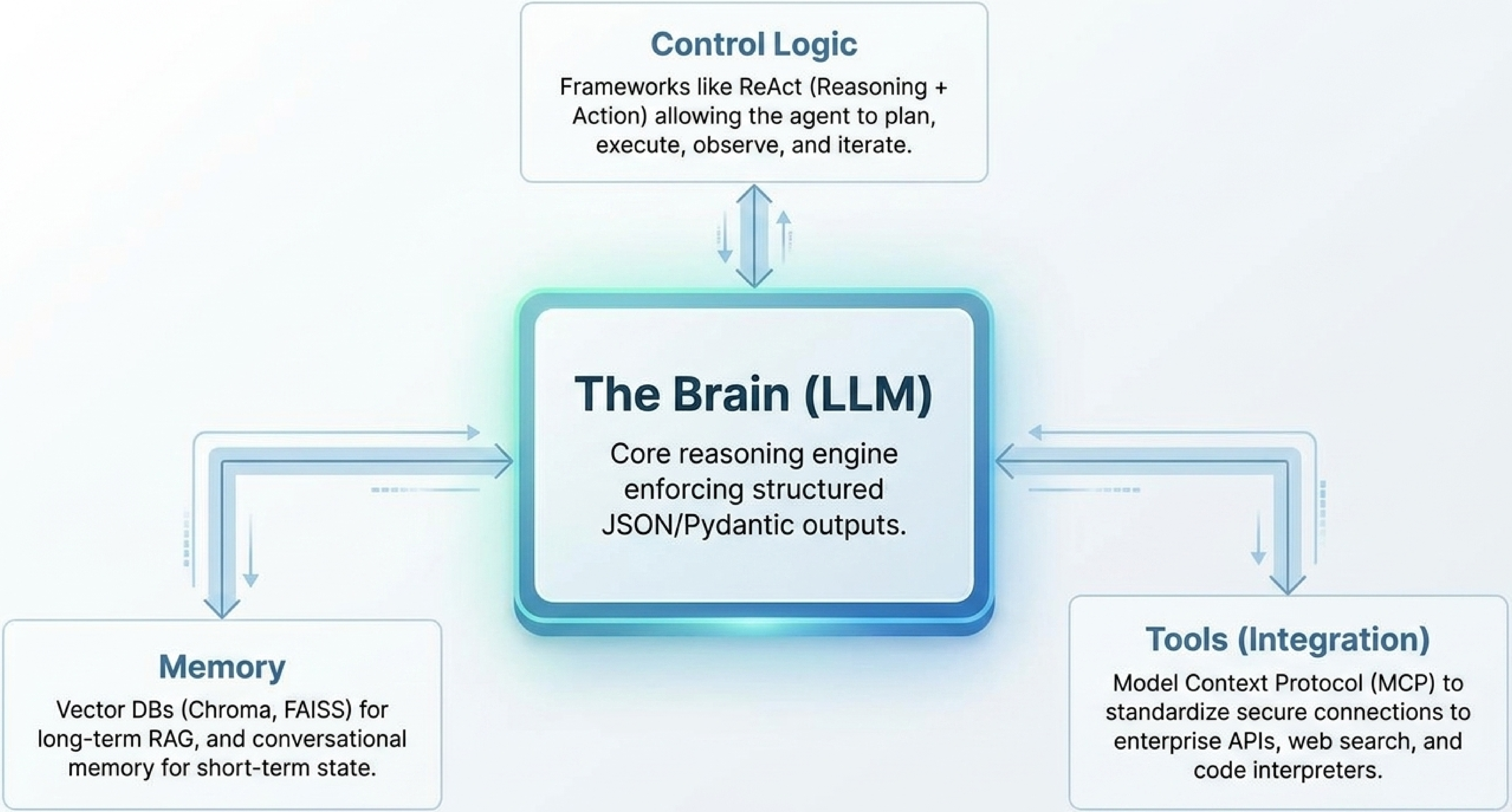


Transformers (2017)

Replaces sequential processing with parallel processing. Self-attention mechanisms weight the importance of every token across the entire sequence simultaneously simultaneously.

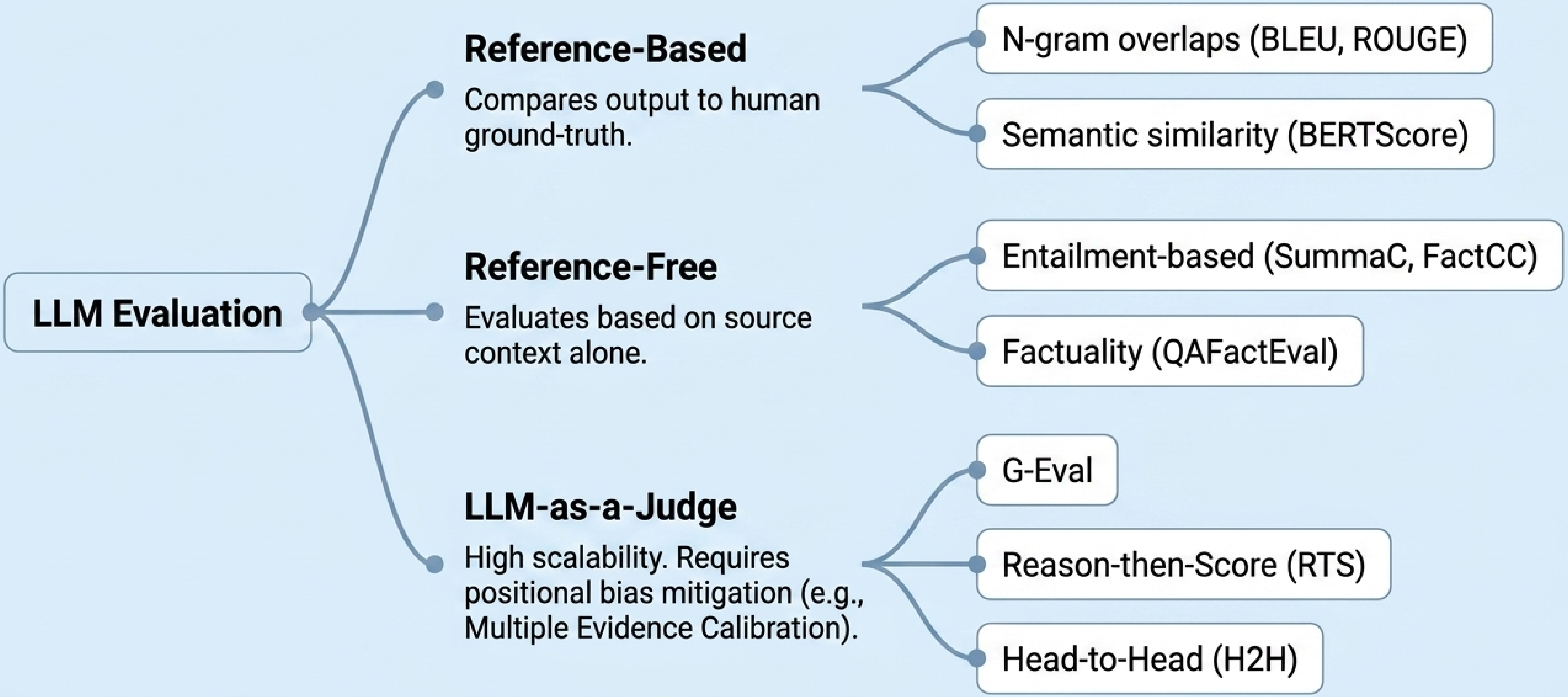
Prompt Engineering Frameworks Comparison.

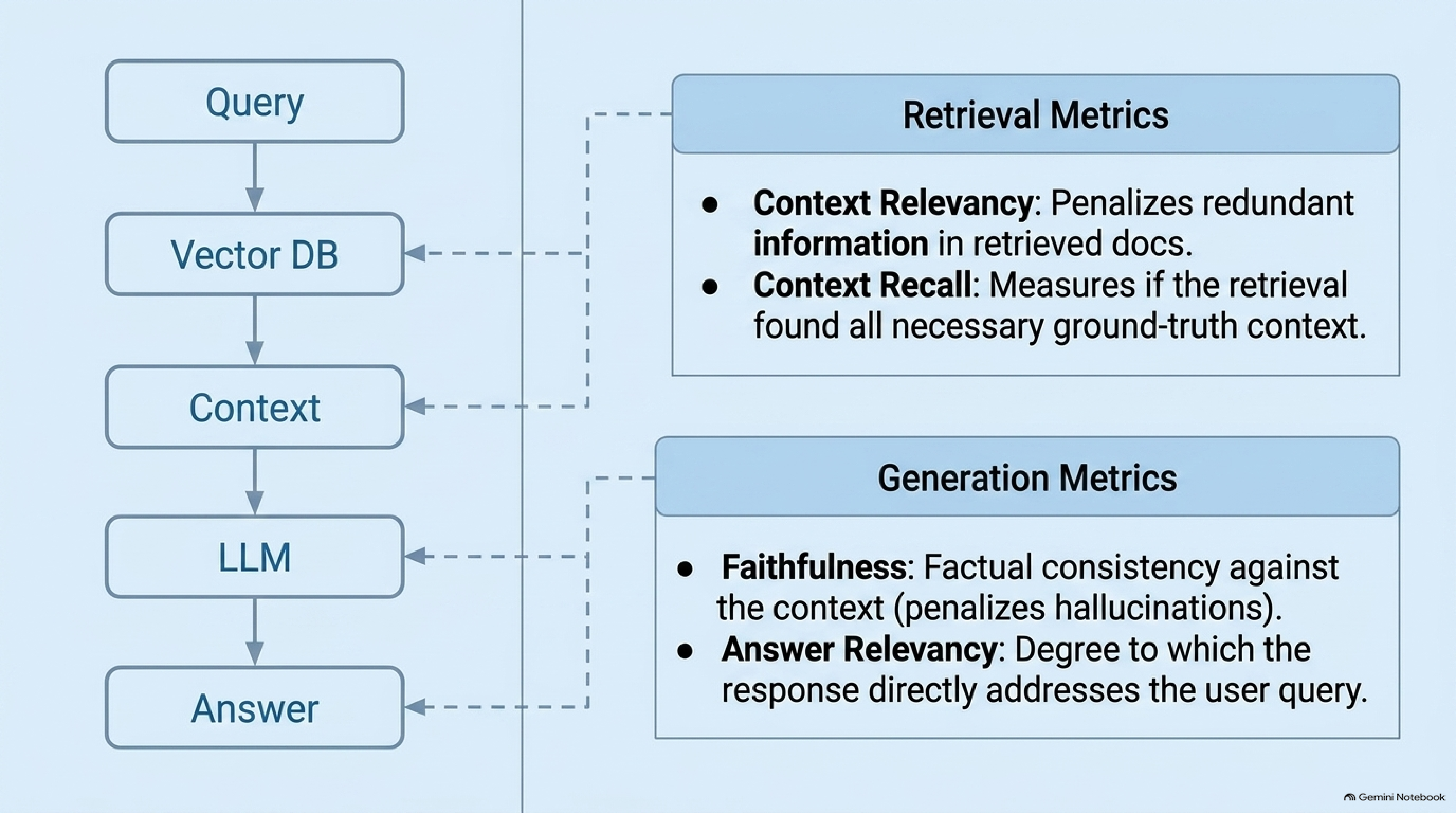
Fast Drafting	Technical / Coding	Strategic / Analytical	Complex / Public-Facing
CARE Context, Action, Result, Example **ERA** Expectation, Role, Action	**RTFC** Role, Task, Format, Constraints	**APE** Action, Purpose, Expectation	**CO-STAR** Context, Objective, Style, Tone, Audience, Response **BRIDGE** Background, Role, Instructions, Details, Guidance, Examples
Best for rapid iteration and professional emails.	Direct, functional, zero conversational filler.	Aligns AI reasoning with broader business objectives.	High-control frameworks for critical tone and formatting.

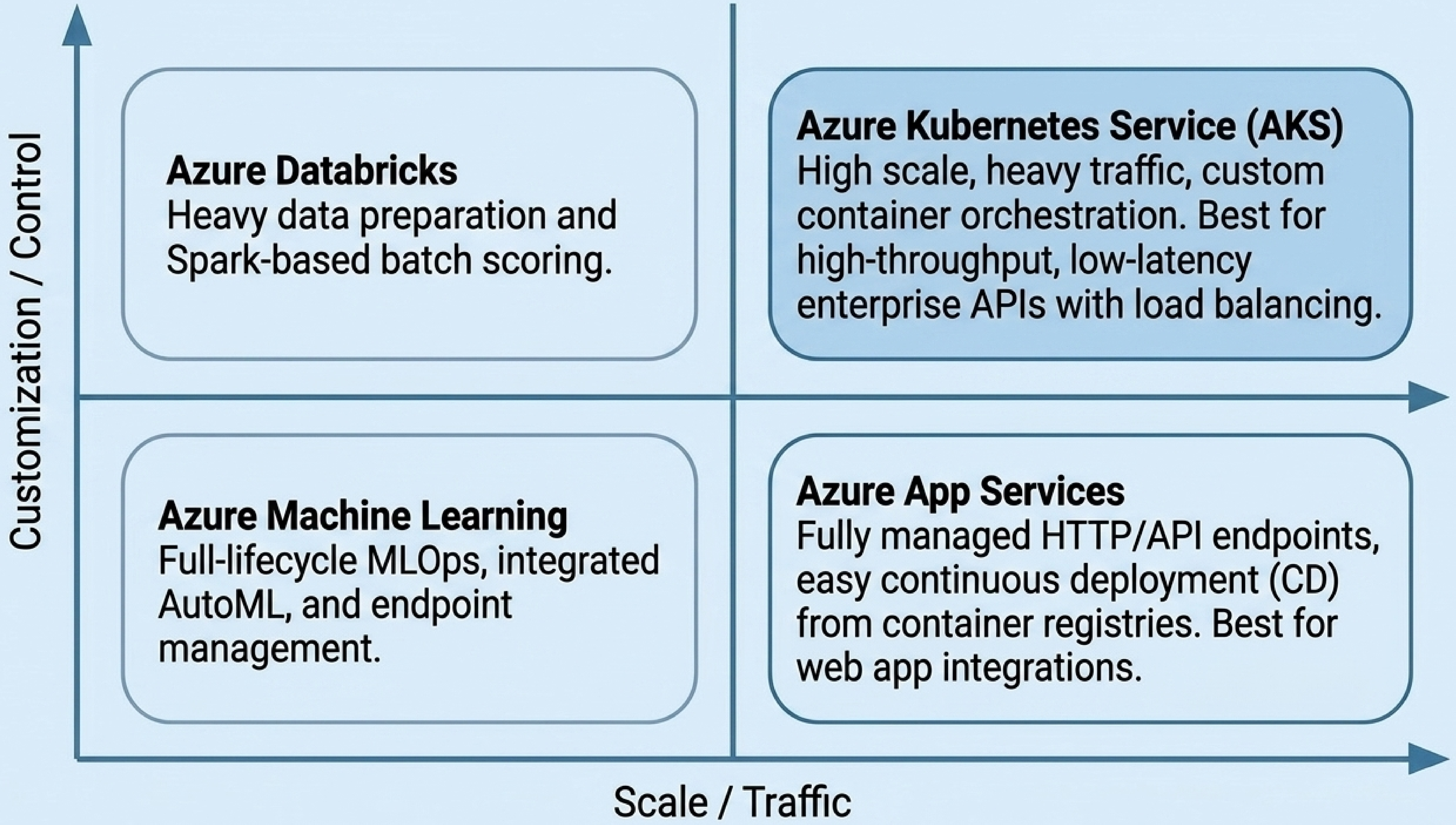


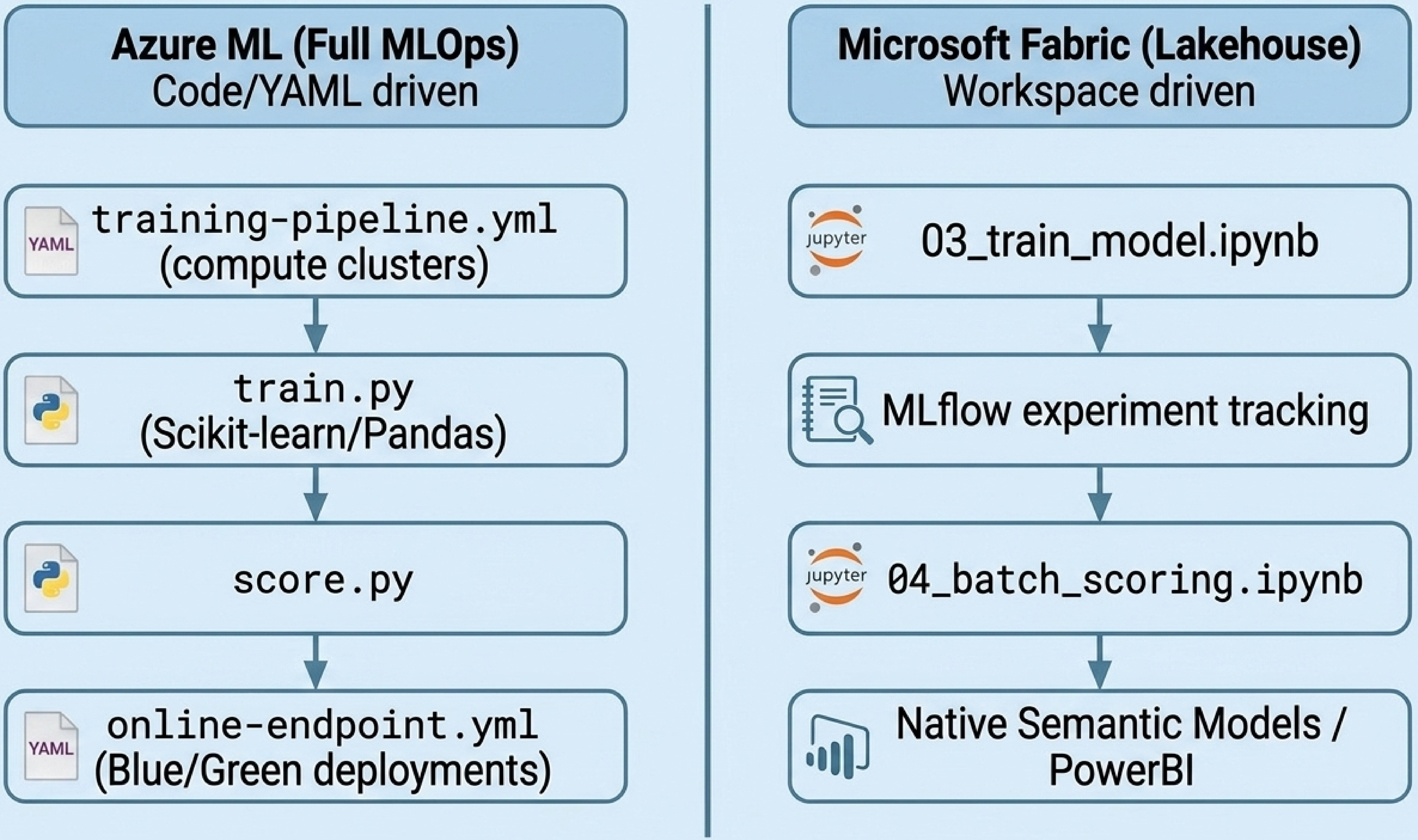
Control Logic
Frameworks like ReAct (Reasoning + Action) allowing the agent to plan, execute, observe, and iterate.
The Brain (LLM)
Core reasoning engine enforcing structured JSON/Pydantic outputs.
Memory
Vector DBs (Chroma, FAISS) for long-term RAG, and conversational memory for short-term state.
Tools (Integration)
Model Context Protocol (MCP) to standardize secure connections to enterprise APIs, web search, and code interpreters.
Gemini Notebook

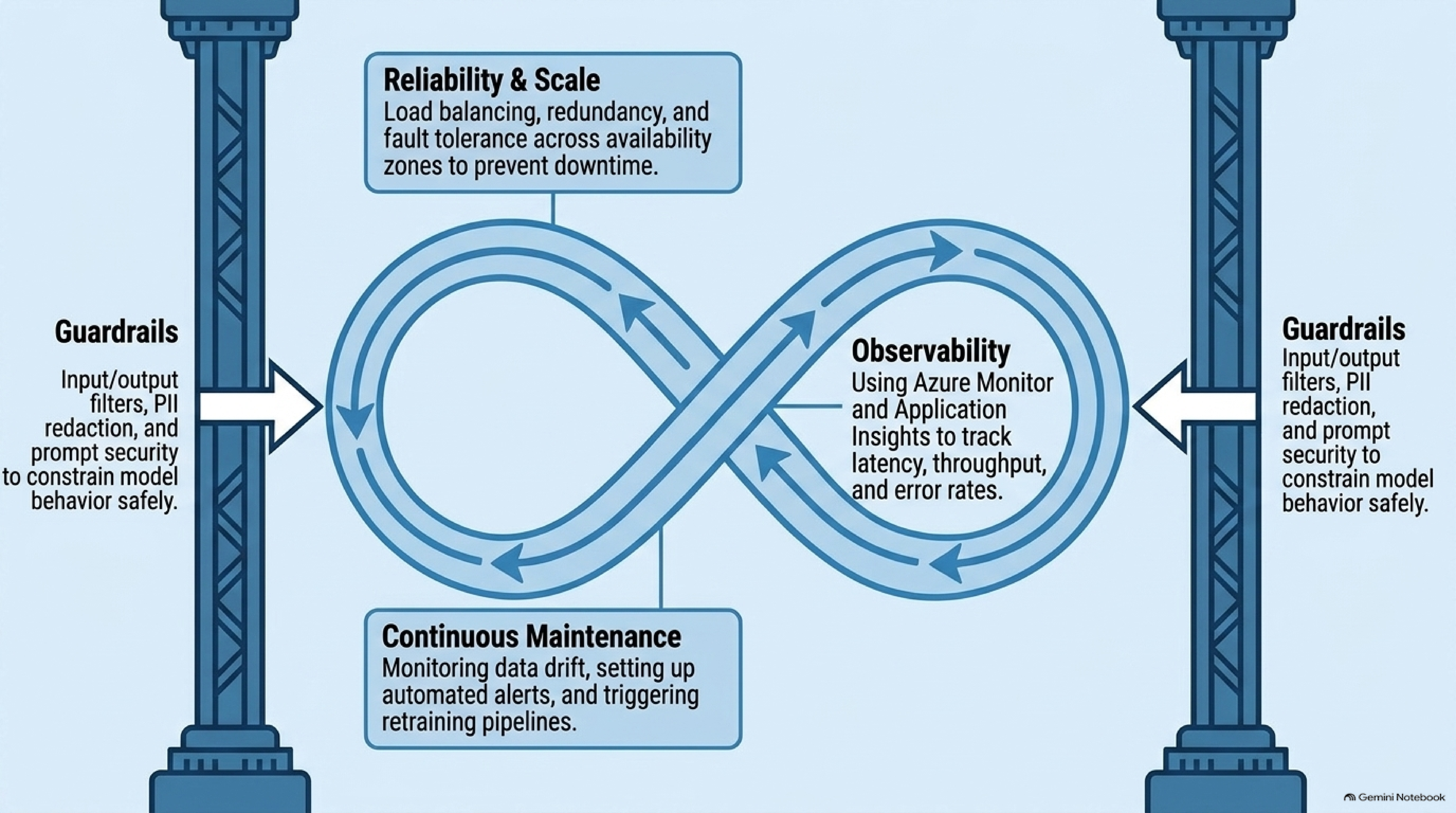
LLM Evaluation Methods Taxonomy.




Query
Vector DB
Context
LLM
Answer
Retrieval Metrics
• **Context Relevancy**: Penalizes redundant **information** in retrieved docs.
• **Context Recall**: Measures if the retrieval found all necessary ground-truth context.
Generation Metrics
• **Faithfulness**: Factual consistency against the context (penalizes hallucinations).
• **Answer Relevancy**: Degree to which the response directly addresses the user query.
Gemini Notebook


Customization / Control
Azure Databricks
Heavy data preparation and Spark-based batch scoring.
Azure Kubernetes Service (AKS)
High scale, heavy traffic, custom container orchestration. Best for high-throughput, low-latency enterprise APIs with load balancing.
Azure Machine Learning
Full-lifecycle MLOps, integrated AutoML, and endpoint management.
Azure App Services
Fully managed HTTP/API endpoints, easy continuous deployment (CD) from container registries. Best for web app integrations.
Scale / Traffic
Gemini Notebook


Azure ML (Full MLOps)
Code/YAML driven
training-pipeline.yml
(compute clusters)
train.py
(Scikit-learn/Pandas)
score.py
online-endpoint.yml
(Blue/Green deployments)
Microsoft Fabric (Lakehouse)
Workspace driven
03_train_model.ipynb
MLflow experiment tracking
04_batch_scoring.ipynb
Native Semantic Models /
PowerBI
Gemini Notebook


Reliability & Scale
Load balancing, redundancy, and fault tolerance across availability zones to prevent downtime.
Guardrails
Input/output filters, PII redaction, and prompt security to constrain model behavior safely.
Observability
Using Azure Monitor and Application Insights to track latency, throughput, and error rates.
Guardrails
Input/output filters, PII redaction, and prompt security to constrain model behavior safely.
Continuous Maintenance
Monitoring data drift, setting up automated alerts, and triggering retraining pipelines.

AI ECOSYSTEM ARCHITECTURE STACK

Layer 4: Safety & LLMOps

Evaluation (RAGAS, DeepEval), Guardrails (NeMo), Observability (LangSmith, MLflow).

Layer 3: The LLM Engine

Foundational Models (OpenAI GPT, Anthropic Claude, Meta Llama).

Layer 2: Orchestration & Logic

Frameworks (LangChain, LlamaIndex, Semantic Kernel), Multi-Agent (CrewAI, AutoGen).

Layer 1: Storage & Embedding

Vector DBs (Chroma, FAISS, Pinecone), Embedding Models (OpenAI, Sentence Transformers).

The modern AI engineer doesn’t just build models; they architect interconnected systems.